

sell sell marketplace updates:

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`SellSell / PickPic Search Improvement Summary (What Happened)

1. Initial Problem

You noticed that AI search was not reliable. Examples:

- “Merced S 400” sometimes worked
 - Color searches like “white car” failed
 - Property searches were inconsistent
 - Gaming searches returned unrelated items
 - Chat searches sometimes got stuck on “**Analyzing request...**” forever
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2. Search Architecture Investigation

The entire search pipeline was analyzed:

- Intent Extractor
- Query Builder
- Tier 1 SQL Search
- Tier 2 Relaxed SQL Search
- Tier 3 Qdrant Semantic Search
- Tier 4 Similar Results

Several issues were found:

Problems Found

1. No typo normalization
 - Merced → Mercedes
 2. Color extracted but not actually filtered
 3. Gaming console searches were too broad
 4. Property searches lacked proper filters
 5. Qdrant failures could hang searches for up to 240 seconds
 6. Location searches were weak because city/area data was generic
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3. New Search Engine Improvements

Implemented:



Deterministic Intent Extraction

Examples:

Query	Extracted
Merced S 400	Mercedes + S400
black mercedes	Mercedes + Black
family SUV under 20000	SUV + Price
sea view chalet	Chalet + Sea View
apartment in Salmiya	Property + Location
gaming console	Gaming Category

4. Added 4-Tier Search Cascade

Tier 1

Exact SQL Match

Example:

- Mercedes
- S400

Tier 2

Relaxed SQL Match

Example:

- Mercedes
- Black

Tier 3

Semantic Qdrant Search

For fuzzy searches.

Tier 4

Similar Listings



Fallback recommendations.

5. Fixed Search Hanging Issue

Root cause:

Qdrant calls had no timeout.

Logs showed:

```
[Cascade] Tier 3 Qdrant failed: fetch failed  
Done in 240393ms
```

Searches could wait 240 seconds.

Fix

Added 5-second timeout to:

- Embedding generation
- Qdrant search
- Similar listing search

Result:

Before:

- Search could hang for 4 minutes

After:

- Search returns within seconds
 - Falls back to SQL results
-

6. Added Debugging & Benchmarking

Created:

```
scripts/search-benchmark.ts
```

Now you can automatically test search quality.

Added logs:

6/8/26 5:47:16 PM



[SearchDebug]
Intent
Tier
Results
Ranking
Scores

7. Ranking Improvements

Added weighted scoring:

Factor	Weight
Brand	40
Model	30
Category	15
Price	10
Location	5

Total:

100 points

8. Inventory Audit

Database inspection found:

Good

- 81 active listings
- Images available
- Vehicles
- Properties
- Electronics

Problems Found

- Some unrealistic property prices

Examples:

151,000 KWD
305,000 KWD



Likely seed/demo data issue.

9. Phase 2 Benchmark

Result:

7 / 8 Passed
88%

Only failure:

white car

Reason:

Color filtering was incomplete.

10. Phase 3 Fixes

Implemented:

Vehicle Detection

Added:

car
cars
truck
vehicle
van
pickup

Now:

white car
black car
red car

correctly map to:

category = vehicles

Color Search Fix

Before:

description contains "red"



Could match unrelated words.

After:

"red color"
"white color"
"black color"

More accurate.

Added Arabic color support too.

Location Search Fix

Now searches:

- Title
- Description
- Property fields

Examples:

Salmiya
Hawally
Kuwait City

work properly.

Property Search Fix

Examples:

luxury villa
sea view chalet

now return the correct property types.

Gaming Search Fix

Now locked to:

PS5
Xbox
Nintendo
Console



instead of general electronics.

11. Final Benchmark

Results:

Core Tests

8 / 8 Passed

Color Tests

3 / 3 Passed

Real User Tests

3 / 3 Passed

Overall

14 / 14 Passed
100%

Current Status

Completed

- Intent extraction
- Search cascade
- Timeout protection
- Ranking improvements
- Color filtering
- Location filtering
- Property filtering
- Gaming filtering
- Benchmarking
- Debug logging

Ready

- Commits prepared
- Deployment plan prepared
- Production verification checklist prepared

Still Recommended

1. Fix unrealistic property prices
2. Add more real inventory
3. Add vehicle color as a structured DB field instead of parsing text
4. Surface `searchTier` in the UI for debugging
5. Build admin dashboard for search analytics



Final Result

The search system evolved from:

Partial matching
Occasional hangs
No proper color filtering
88% benchmark score

to:

Deterministic intent extraction
4-tier search engine
5-second fail-safe timeouts
Color-aware search
Location-aware search
Property-aware search
Gaming-aware search
14/14 benchmark score
100% pass rate
Ready for production deployment